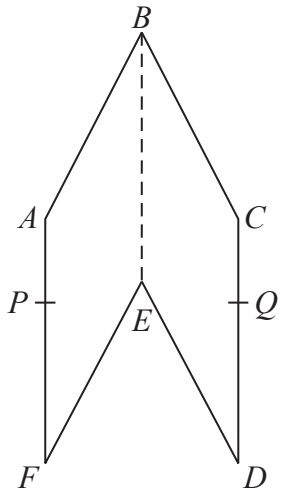
22 The diagram shows a hexagon $ABCDEF$.



$ABEF$ and $CBED$ are congruent parallelograms where $AB = BC = x$ cm.
P is the point on AF and Q is the point on CD such that $BP = BQ = 10$ cm.

Given that angle $ABC = 30°$,

prove that $\cos PBQ = 1 - \dfrac{(2 - \sqrt{3})}{200}x^2$

Congruent parallelograms give symmetry about BE;

with BP = BQ, AP = CQ, so PQ = AC.

$\cos 30° = \dfrac{\sqrt{3}}{2}$

1 mark
Exact cosine of 30°

$AC^2 = x^2 + x^2 - 2 \times x \times x \times \cos 30°$

$= 2x^2 - (\sqrt{3})x^2 = (2 - \sqrt{3})x^2$

1 mark
Apply the cosine rule

In triangle PBQ, let θ = angle PBQ.

$PQ^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos\theta$

$\cos\theta = \dfrac{200 - PQ^2}{200}$

1 mark
Rearrange the cosine rule

$= \dfrac{200 - (2 - \sqrt{3})x^2}{200}$

1 mark
Substitute $PQ^2 = AC^2$

$= 1 - \dfrac{(2 - \sqrt{3})x^2}{200}$

1 mark
Complete the required proof

(Total for Question 22 is 5 marks)